EE2005 – ELECTRICAL DEVICES AND MACHINES

TUTORIAL 3 – MAGNETIC LOSSES AND ELECTROMECHANICAL ENERGY CONVERSION

1. The core losses in a certain specimen of iron were found to be 4000 W at a frequency of 50 Hz. When the supply frequency is changed to 30 Hz maintaining the flux density unchanged, the core losses were found to be 1800 W. Calculate the eddy current loss at the frequency of 50 Hz.

   [**Ans: 2500 W**]

2. An inductor core is designed to operate at 120 V at 60 Hz. Estimate the effect on the hysteresis and eddy current losses of operating at 150 V at 50 Hz. Assume that the Steinmetz index $n = 1.6$.

   [**Ans: Losses increase by 0.59 and 0.56, respectively**.]

3. The electromagnet shown in Figure 1 is used to lift a steel bar. The system is excited by a coil of 2500 turns wound over the semi-circular part of the magnetic core. The cross section of the magnetic core is 20 mm x 40 mm throughout the structure. The air gap 'g' is 10 mm. Assume that the permeabilities of the magnetic core and the steel bar are infinitely high. The permeability of free space $\mu_o = 4\pi \times 10^{-7}$ H/m.

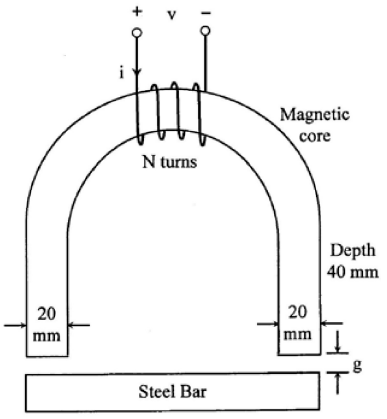


Figure 1

(a) Find the coil current required to produce a flux density of 1.25 T in the air gap.
(b) Determine the energy stored in the magnetic system.
(c) Calculate the force acting on the steel bar.

If the air gap 'g' is reduced to 5 mm, determine the coil current required to produce the same force as in part (c) above.

[**Ans: (a) 7.96 A; (b) 9.95 J; (c) 994.72 N; 3.98 A**]

4. The magnetic circuit, as shown in Figure 2 is excited by a 100-turn coil wound over the central limb. The circuit has a uniform cross-sectional area of 1 cm^2 and the mean lengths of the various sections are $l_s = 15.5$ cm and $l_c = 5.5$ cm. The length of the iron bar is 11 cm. The air gap length is x m. Determine the current in the coil that is necessary to keep the movable part (iron bar) suspended at 1 cm from the immovable part (magnetic structure). What is the energy stored in the system? The relative permeability of both the magnetic material and the iron bar is 2000 and the density of the iron bar is 7.85 g/cm^3, respectively. The acceleration due to gravity g = 9.81 m/s^2.

[Ans: 14.28 A, 8.5 mJ]

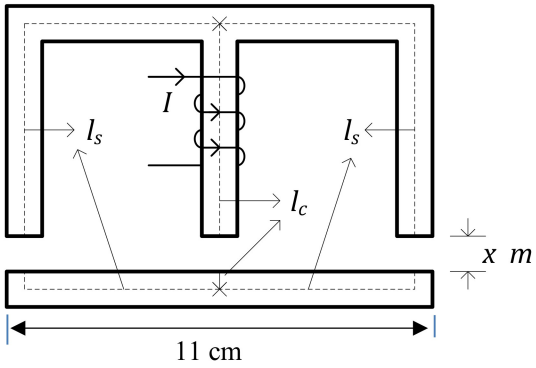


Figure 2

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TUTORIAL 4 - TRANSFORMER FUNDAMENTALS AND APPROXIMATE EQUIVALENT CIRCUITS

1. A single-phase, 1-kVA, 250/100-V, 60-Hz transformer has 120 turns on the primary winding. Calculate (a) the turns ratio, (b) the number of turns on the secondary winding, and (c) the rated currents of the two windings. Find the maximum flux in the core when the transformer is connected to a 250-V, 60-Hz source. If the 250-V terminals were connected to a 220-V, 50-Hz supply, determine the percent change in the flux density in the core.

   [**Ans: (a) 2.5; (b) 48 turns; (c) 4 A, 10 A; 7.82 mWb, 5.6%**]

2. Consider a 200 kVA, 22000/2200-V, 50 Hz single-phase transformer with the following parameters:

   $R_1 = 5.0\ \Omega$, $X_1 = 30\ \Omega$, $R_2 = 0.03\ \Omega$, $X_2 = 0.1\ \Omega$,
   $R_{c1} = 15\ k\Omega$, $X_{m1} = 7.5\ k\Omega$

   Draw the approximate equivalent circuits of the transformer referred to the high-voltage side and to the low voltage side, showing the proper numerical values of the parameters in the circuits.

3. The nameplate of a transformer reads: single-phase, 100 kVA, 11000/2200 V, 60 Hz.

   (a) Calculate the full-load (rated) currents in the high-voltage and low-voltage sides.

   [**Ans: 9.09 A, 45.45 A**]

   (b) The transformer has the following parameters:

   $R_1 = 6.0\ \Omega$, $X_1 = 30.16\ \Omega$, $R_2 = 0.28\ \Omega$, $X_2 = 1.21\ \Omega$,
   $R_{c1} = 125\ k\Omega$, $X_{m1} = 60.32\ \text{k}\Omega$

   Draw the approximate equivalent circuit of the transformer referred to the high-voltage side and use it to carry out the following calculations.

   (c) Calculate the:
   (i) No-load current and no-load power factor,
   (ii) No-load power loss (i.e., core loss), and
   (iii) Full-load copper loss.

   [**Ans: (i) 0.2 A, 0.44 (lag); (ii) 968 W; (iii) 1074 W**]

   (d) Calculate the input voltage and the input current while supplying full load at 0.85 lagging power factor at rated voltage. Take the rated secondary voltage as the reference.

   [**Ans: 11397 V, 9.26 A]**

   (e) Repeat part (d) above using the approximate equivalent circuit of the transformer referred to the low-voltage side.

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TUTORIAL 5 – TRANSFORMER EFFICENCY AND TESTING

1. The maximum efficiency of a 100-kVA transformer is 98.4% and occurs at 90 per-cent of full load, at unity power factor. Calculate the efficiency of the transformer at unity power factor at (a) full load and (b) 50 percent of full load. [Hint: Use the expression for efficiency and note the condition for maximum efficiency and the core losses are assumed constant.]

   [**Ans: 98.39; 98.12**]

2. In a 50-kVA, 11-kV/400-V, single-phase transformer, the core and copper losses are 500 W and 600 W, respectively under rated conditions. Calculate (a) the efficiency at unity power factor at full load, (b) the load for maximum efficiency (Hint: kVA at maximum efficiency = rated kVA x $\sqrt{\frac{P_{core}}{P_{cu(fl)}}}$) and (c) the iron and copper losses for this load.

   [**Ans: 97.85 %; 45.64 kVA; 500 W; 500 W**]

3. A 500-kVA, 30/10-kV, 50-Hz, single-phase transformer has the high-voltage winding resistance of 18 Ω and the low-voltage winding resistance of 4 Ω. Use the approximate equivalent circuit referred to the low voltage side to do the calculations.

   (a) Determine the total copper loss of the transformer at full load.

   [Ans: 15 kW]

   (b) If the total core loss of the transformer is 10 kW, estimate the efficiency of the transformer at half load at 0.85 lagging power factor.

   [**Ans: 93.92%**]

   (c) What is the maximum efficiency of the transformer at 0.85 lagging power factor?

   [**Ans: 94.55%**]

4. The 75-kVA, 4600/230-V, 50-Hz transformer gave the following test results.

   Open-Circuit Test (with 4600 V side open)

   V_{oc} = 230 V, I_{oc} = 13.04 A, P_{oc} = 521 W

   Short-Circuit Test (with 230 V side shorted)

   V_{sc} = 160.8 V, I_{sc} = 16.3 A, P_{sc} = 1200 W

   (a) Determine the parameters of the approximate equivalent circuit of the transformer referred to the primary side.

   [**Ans: R_{c1} = 40.616 kΩ, X_{m1} = 7164 Ω, R_{e1} = 4.517 Ω, X_{e1} = 8.77 Ω**]

   (b) Draw the approximate equivalent circuit referred to the low-voltage side showing all the circuit parameter values.

   (c) Use the circuit in (b) to calculate the voltage regulation and the efficiency of the transformer when it delivers full load at 0.85 lagging power factor at rated voltage.

   [**Ans: 3.01%, 97.32%**]

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TUTORIAL 6 - AUTOTRANSFORMERS AND THREE-PHASE TRANSFORMER BANKS

1. A 1000/200-V, 25-kVA, 50-Hz transformer is connected as an autotransformer to yield a transformation ratio of 1000/1200. Calculate its kVA rating. Calculate also the currents in the two windings when the autotransformer is fully loaded.

   [**Ans:150 kVA; 125 A; 25 A**]

2. The primary and secondary voltages of an autotransformer are 440 V and 360 V respectively. If the secondary current is 80 A, determine the primary current. Calculate the power transferred to the load inductively and conductively. Also, determine the kVA rating of the autotransformer.

   [**Ans: 65.45 A; 5.235 kVA; 23.56 kVA; 28.8 kVA**]

3. A 10-kVA, 2400/240-V, 50-Hz distribution transformer is to be used as a step-up autotransformer. The input voltage is 2400 V. Determine the current ratings of the two windings and the maximum power that can be delivered with this connection. Calculate the amount of power transferred by (a) conduction, and (b) magnetic induction. Ignore losses.

   [**Ans: 4.167 A, 41.67 A, 110 kVA, 100 kVA, 10 kVA**]

4. A three-phase transformer bank is to handle 500 kVA and have a 34.5/11-kV voltage ratio. Find the rating of each individual transformer in the bank (high voltage, low voltage, and apparent power) if the transformer bank is connected in the following arrangements: (a) Y-Y, (b) Y-Δ, (c) Δ-Y, (d) Δ-Δ.
   [**Ans: Y-Y 19.9 kV 6.35 kV 167 kVA; Y-Δ 19.9 kV 11.0 kV 167 kVA; Δ-Y 34.5 kV 6.35 kV 167 kVA; Δ-Δ 34.5 kV 11.0 kV 167 kVA**]

5. An ideal three-phase step down transformer, connected Δ-Y delivers power to a balanced three-phase load of 120 kVA at 0.8 power factor. The input line voltage is 11 kV and the turns- ratio of the transformer, phase to phase is 10. Determine the line voltage, line currents, phase voltages and phase currents on both the primary and the secondary sides. What is the power (kW) consumed by the load?

   [**Ans: V_{LY} = 1.9 kV; $V_{L\Delta}$ = 11 kV; I_{LY} = 36.4 A; $I_{L\Delta}$ = 6.3 A; $I_{\phi\Delta}$ = 3.64 A; 96 kW**]

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TUTORIAL 7 - THREE-PHASE TRANSFORMERS

1. A three-phase transformer is assembled by connecting three 480-VA, 240/80-V, single-phase transformers. The constants for each transformer are R_1 =12.6 Ω, X_1 = 14.4 Ω, R_2 = 1.4 Ω, X_2 = 1.6 Ω, R_{c1} = 5.76 kΩ, and X_{m1} = 0.456 kΩ. If the transformer bank is connected in the Y-Y, Y-Δ, Δ-Y and Δ-Δ configurations, draw the per-phase Y-Y approximate equivalent circuit embodying an ideal transformer for each configuration. Also, draw the per-phase Y-Y approximate equivalent circuit referred to the primary side and find Z_{e1} for each configuration.

   [**Ans**: Y-Y: $R_1 = 12.6\,\Omega$, $X_1 = 14.4\,\Omega$, $R_2 = 1.4\,\Omega$, $X_2 = 1.6\,\Omega$ $R_{c1} = 5.76\,k\Omega$, $X_{m1} = 0.456\,k\Omega$; $Z_{Ye1}|_{\text{Y-Y}} = 25.2 + j28.8\,\Omega$.
   Y-Δ: $R_1 = 12.6\,\Omega$, $X_1 = 14.4\,\Omega$, $R_2 = \frac{1.4}{3}\,\Omega$, $X_2 = \frac{1.6}{3}\,\Omega$ $R_{c1} = 5.76\,k\Omega$, $X_{m1} = 0.456\,k\Omega$; $Z_{Ye1}|_{\text{Y-}\Delta} = 25.2 + j28.8\,\Omega$.
   Δ-Y: $R_1 = \frac{12.6}{3}\,\Omega$, $X_1 = \frac{14.4}{3}\,\Omega$, $R_2 = 1.4\,\Omega$, $X_2 = 1.6\,\Omega$, $R_{c1} = \frac{5.76}{3}\,k\Omega$, $X_{m1} = \frac{0.456}{3}\,k\Omega$; $Z_{Ye1}|_{\Delta\text{-Y}} = 8.4 + j9.6\,\Omega$.
   Δ-Δ: $R_1 = \frac{12.6}{3}\,\Omega$, $X_1 = \frac{14.4}{3}\,\Omega$, $R_2 = \frac{1.4}{3}\,\Omega$, $X_2 = \frac{1.6}{3}\,\Omega$, $R_{c1} = \frac{5.76}{3}\,k\Omega$, $X_{m1} = \frac{0.456}{3}\,k\Omega$; $Z_{Ye1}|_{\Delta-\Delta} = 8.4 + j9.6\,\Omega$.]

2. Three identical single-phase transformers, each rated 50 kVA, 2300/230 V, 50 Hz, are connected to form a three-phase Δ-Y transformer bank. The total effective impedance of each single-phase transformer referred to its high-voltage side is (1.33+j4.0) Ω. Ignore the no-load current. Using the per-phase Y-Y equivalent circuit referred to the primary side, determine the input current and input voltage of the three-phase transformer bank when it delivers full load at 0.9 lagging power factor at rated voltage. You may use the rated phase secondary voltage as the reference.

   [**Ans: 37.65 A, 2365 V**]

3. Suppose the three identical single-phase transformers in Question 2 are now connected to form a three-phase Y-Δ transformer bank. Using the per-phase Y-Y equivalent circuit referred to the primary side, determine the input current and input voltage of the three-phase transformer bank when it delivers full load at 0.9 lagging power factor at rated voltage.

   [**Ans: 21.74 A, 4096 V**]

4. A 100-kVA, 2300/460-V, 50-Hz, Y-Δ-connected, three-phase transformer has the total effective impedance of (0.045+j0.16) Ω/phase referred to its low-voltage side. Ignore the no-load current. The transformer is connected to a three-phase source through three-phase feeders. The impedance of each feeder is (0.5+j1.5) Ω. The transformer delivers full load at 460 V and 0.85 lagging power factor.

   (a) Draw the per-phase Y-Y equivalent circuit of the system referred to the primary side.
   (b) Determine the sending end voltage of the three-phase source.
   (c) Calculate the complex power input at the sending end of the feeder.

   [**Ans:** (b) **2398 V, (c) (86.6+j58) kVA**]